



WEEK 4 – REPORTING AND DESIGN DOCUMENT

End-to-End Salesforce CRM Transformation for Customer Operations and Service Management

7.1 Report Definition

To satisfy the core business objective of enabling real-time management visibility (**Milestone 1**) and measuring the operational performance improvements over the legacy spreadsheet system (**Milestone 2**), GreenGrid Utilities leverages native Salesforce reporting engines.

Reports are structured using specific report types that cross standard and custom object boundaries, ensuring clean data aggregation without duplications.

7.1.1 Core Operational Report Specifications

a. Report 1: Average Handling Time (AHT) and Case Triage Performance

- **Report Type:** Cases (Standard Report Type)
- **Format:** Summary Report (Grouped by Owner and Priority)
- **Core Metrics:** Average Age (Hours/Minutes), Total Case Count, and Breakdown of Case Status.
- **Business Justification:** Directly monitors **BR1** (25% reduction in AHT). It surfaces how quickly agents log, handle, and hand off incoming utility outage calls within the Salesforce Service Console compared to the legacy 5-minute spreadsheet bottleneck (**B1**).

b. Report 2: Field Visit Resolution Efficiency

- **Report Type:** Cases with Field Visits (Custom Report Type based on Case parent and Field_Visit__c child relationship)
- **Format:** Matrix Report (Rows grouped by Field_Visit__c.Owner / Field Technician; Columns grouped by Field_Visit__c.Repair_Status__c)
- **Core Metrics:** Record Count (Total Dispatches), Field Visit Success Rate (Ratio of *Resolved* vs. *Failed*), and Average Turnaround Time.

- **Business Justification:** Measures field operational performance (**BR3** - 15% improvement in first-visit fix rate) and tracks the transition out of informal, untracked messaging applications (**B2**).

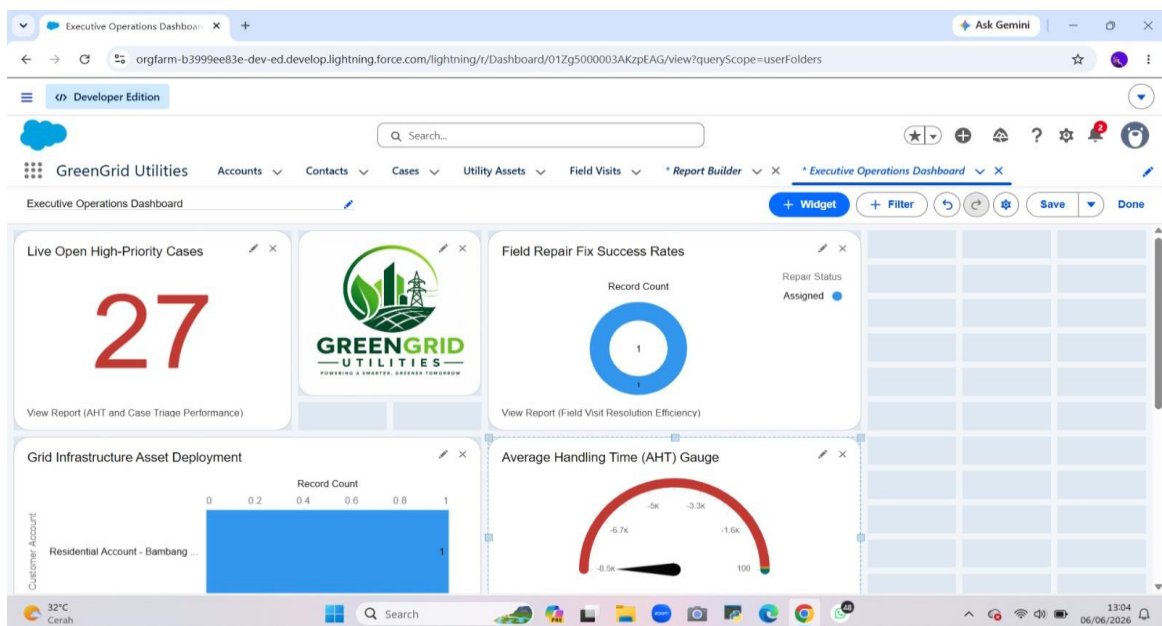
c. Report 3: High-Priority Grid Outage Trends

- **Report Type:** Cases with Utility Assets (Custom Report Type based on Case parent and Utility_Asset__c child relationship)
- **Format:** Summary Report (Grouped by Utility_Asset__c.AssetStatus__c and Asset Installation Address/Region)
- **Core Metrics:** Total High-Priority Cases, Outage Duration Tracking, Asset Failure Frequency.
- **Business Justification:** Provides Executive Leadership with instant visibility into which critical grid hardware infrastructure elements are experiencing recurring service disruptions.

7.2 Dashboard Design

Dashboards act as the unified command center for GreenGrid Utilities' leadership tier (**Apex Tier** in the Role Hierarchy). They translate rows of raw report data into real-time, scannable visual performance components.

7.2.1 Executive Operations Dashboard Blueprint



Component 1 (Top Left) – Open Crisis Metric

Large single-number metric component displaying the active count of High-Priority Cases where Status = New or Working. It updates instantly as **Flow 1 (Dispatch Engine)** executes.

Component 2 (Top Center) – Service Console Speed Gauge

A visual gauge component tracking Average Case Age. The green zone denotes optimal sub-2-minute handling speed, directly proving the resolution of spreadsheet bottleneck **B1**.

Component 3 (Top Right) – First-Visit Fix Ratio Donut

A donut chart showing the proportions of Field_Visit__c.Repair_Status__c (Assigned, In Progress, Resolved, Failed). This measures the elimination of paper log losses (**B3**).

Component 4 (Bottom Main) – Grid Outage Distribution Bar Chart

A vertical bar chart tracking Case counts grouped by Asset Installation Address. This enables executive planning for physical infrastructure reinforcement.

7.3 Data Visualization Strategy

To uphold the performance and configuration standards established in **NFR3** (Dashboards must load within a 5-second window), GreenGrid Utilities enforces strict data visibility rules and dashboard access guardrails.

7.3.1 Role Hierarchy Integration & Dashboard View Context

- Dashboards are configured to run under a specific administrative user context ("View Dashboard As: Specified User" set to an Executive Manager).
- This configuration allows the system to pre-cache complex multi-object roll-ups across the entire organization. It bypasses the performance lag caused by real-time role-based sharing recalculations during a live presentation, satisfying **NFR3** load-time parameters.

7.3.2 Foldering and Security Governance

- Reports and dashboards are housed in a restricted folder structural environment named Executive Operational Analytics.
- **Access Control:** The folder access profile is granted as **Viewer** for the Customer Support Team and Field Technicians role groups to eliminate workspace clutter, and granted as **Manage/Editor** for Executive Management and IT/System Administrators.

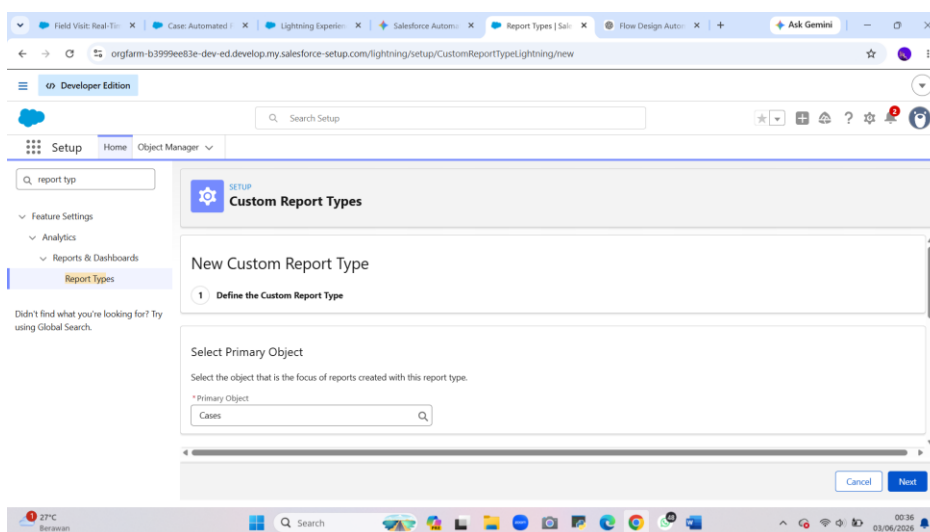
7.4 Salesforce Implementation and Configuration Steps

Follow these precise step-by-step instructions to define custom report structures, build analytical components, construct the executive dashboard layout, and secure the operational folders.

Step 1: Create Custom Report Types for Multi-Object Relationships

Before building the reports, custom frameworks must be deployed to link your custom entities (Utility_Asset__c and Field_Visit__c) with standard objects.

1. Navigate to **Setup** by clicking the gear icon in the top-right corner.
2. In the left-hand **Quick Find** box, type Report Types and select **Custom Report Types**.
3. If an introductory screen appears, click **Continue**.
4. Click the **New Custom Report Type** button.
5. Configure the Report Type definition settings precisely:
 - **Primary Object:** Select Cases.



- **Report Type Label:** Type Cases with Field Visits.

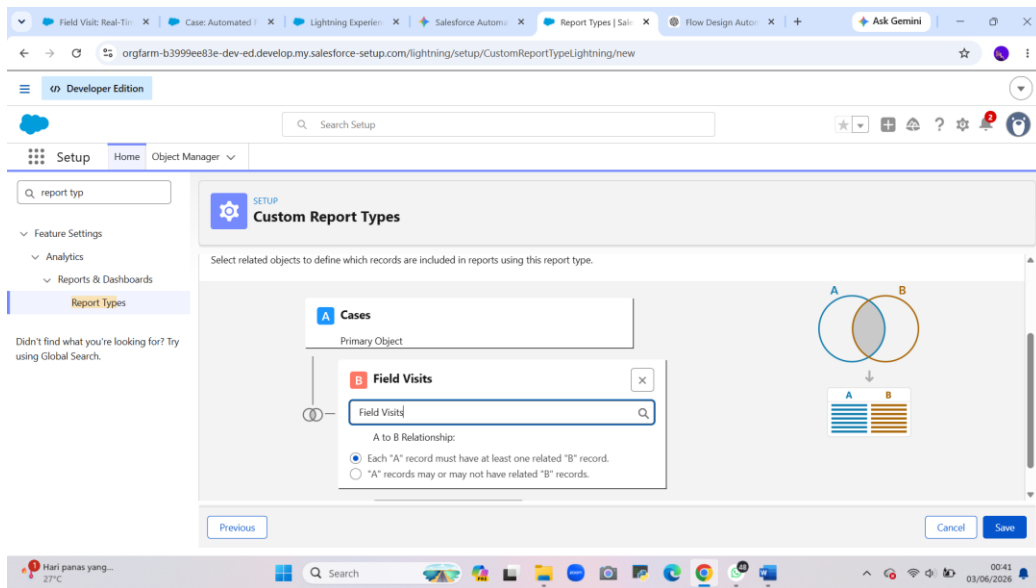
- **Report Type Name:** Auto-populates as Cases_with_Field_Visits.
- **Description:** Type Operational report linking customer service tickets to field crew dispatch actions.
- **Store in Category:** Select Customer Support Cases.
- **Deployment Status:** Select Deployed.

The screenshot shows the Salesforce Setup interface in a web browser. The browser tabs include 'Field Visit: Real-Tim...', 'Case: Automated F...', 'Lightning Experien...', 'Salesforce Autom...', 'Report Types | Sale...', 'Flow Design Auto...', and 'Ask Gemini'. The address bar shows the URL: 'orgfarm-b3999ee83e-dev-ed.develop.my.salesforce-setup.com/lightning/setup/CustomReportTypeLightning/new'. The page title is 'Developer Edition'. The left sidebar shows the 'Setup' menu with 'Home' and 'Object Manager' options. The 'Report Types' section is selected. The main content area is titled 'Custom Report Types' and contains a 'Details' section with the following fields:

- * Display Label: Cases with Field Visits
- * API Name: Cases_with_Field_Visits
- * Description: Operational report linking customer service tickets to field crew dispatch actions.
- Note: Description will be visible to users who create reports.
- * Store in Category: Customer Support Reports

At the bottom right of the form, there are 'Cancel' and 'Next' buttons. The Windows taskbar at the bottom shows the date and time as 03/06/2026 00:38.

6. Click **Next**.
7. On the **Define Object Relationships** canvas layout screen, click the empty box below **A (Cases)** to add a child object.
8. Select **Field Visits (Field_Visit__c)** from the dropdown list. Leave the relationship setting at the default option: Each "A" record must have at least one related "B" record.

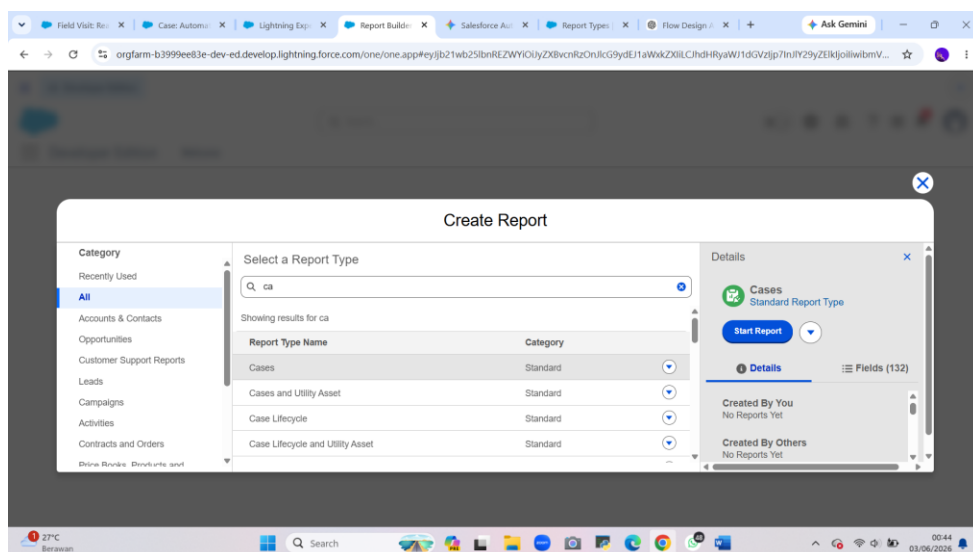


9. Click **Save**.

10. (Optional Repeat for Assets) Click **New Custom Report Type** again. Create a type with **Cases** as Primary, link **Utility Assets (Utility_Asset__c)** as object B, label it Cases with Utility Assets, set it to **Deployed**, and click **Save**.

Step 2: Build Report 1 — Average Handling Time (AHT) Analytics

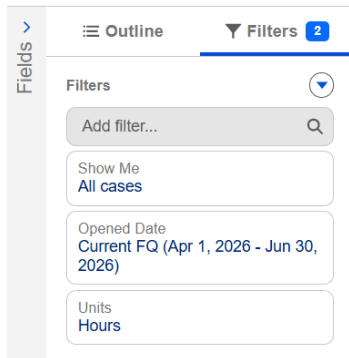
1. Click the **App Launcher** icon and type Reports.
2. Click on **Reports**.
3. Click **New Report**.
4. Search for and select the standard **Cases** report type.
5. Click **Start Report**.



6. Enable **Update Preview Automatically**.

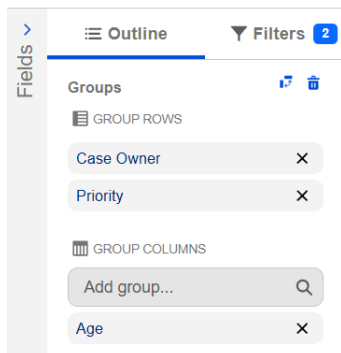
7. Under **Filters**:

- Change **Show Me** from *My Cases* to **All Cases**.
- Set **Opened Date** to **Current FQ** (or All Time for testing).



8. Under **Outline**:

- Group Rows by **Case Owner**.
- Add second grouping by **Priority**.



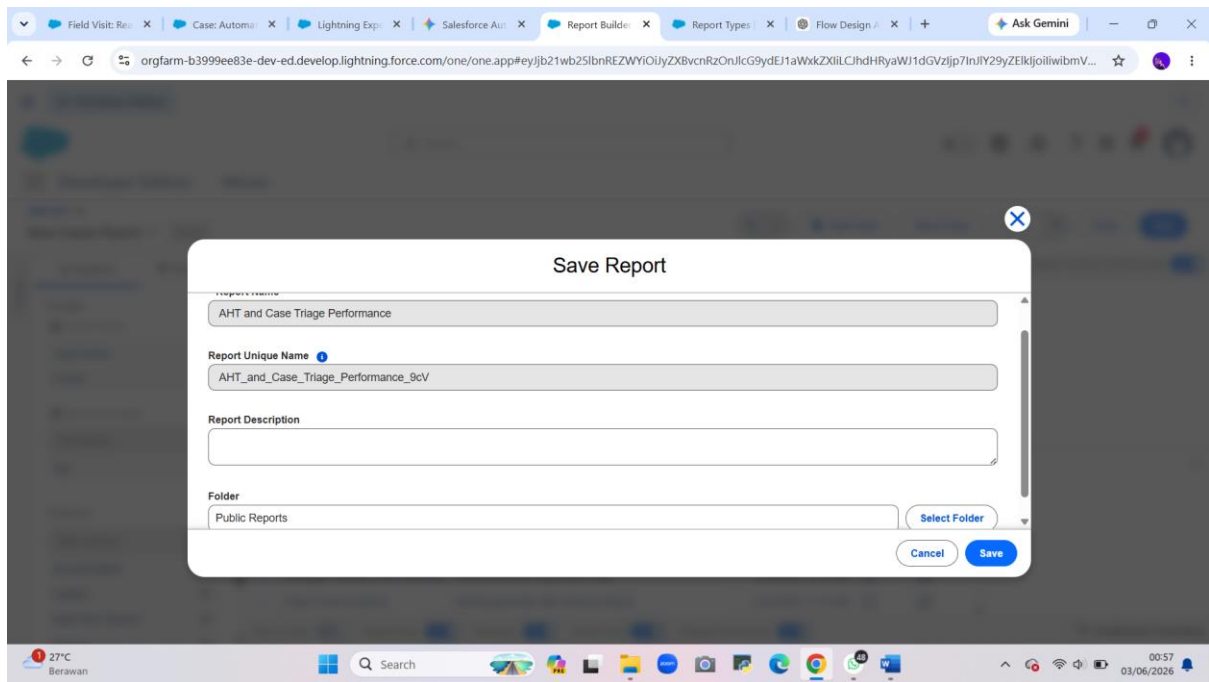
9. Add the **Age** column and remove unnecessary columns.

10. Summarize the **Age** field using **Average**.

11. Click **Save & Run**.

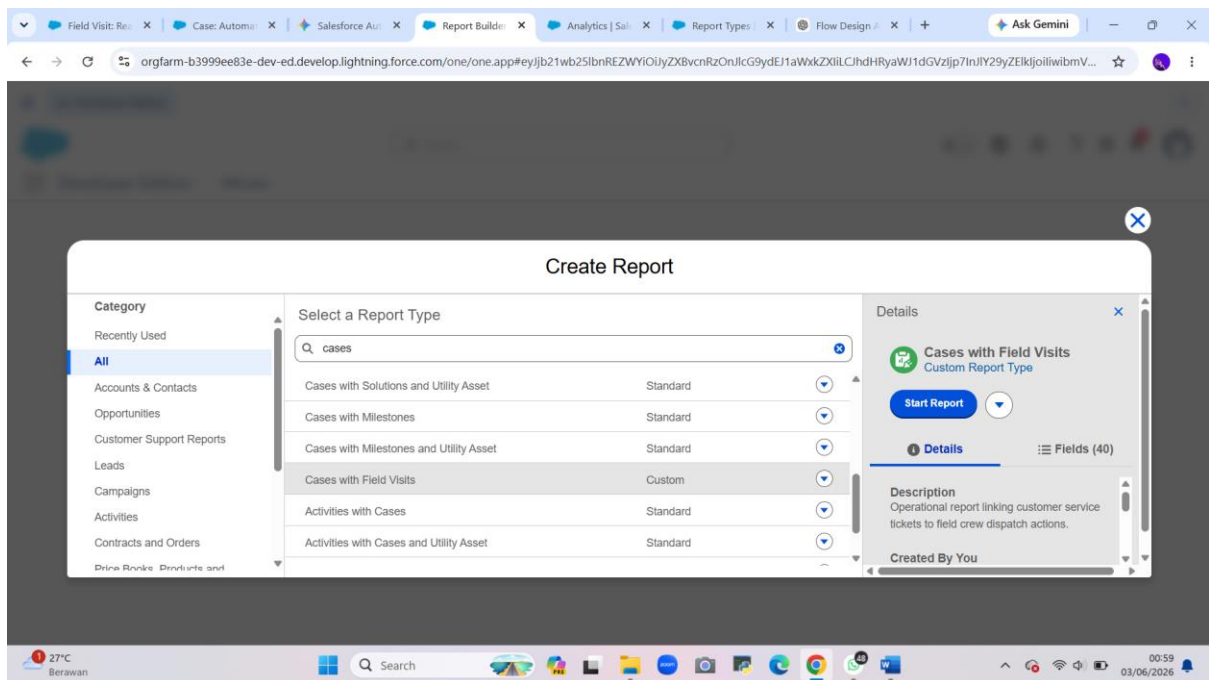
12. Save as:

- Report Name: AHT and Case Triage Performance
- Unique Name: AHT_and_Case_Triage_Performance



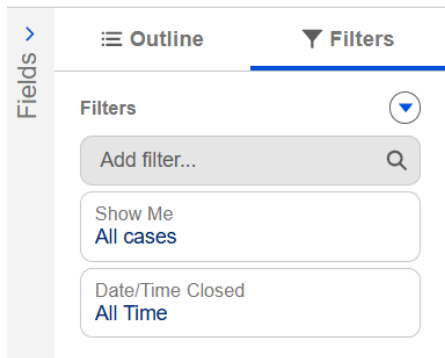
Step 3: Build Report 2 — Field Visit Resolution Efficiency

1. Click **New Report**.
2. Select **Cases with Field Visits**.
3. Click **Start Report**.



4. Under **Filters**:
 - Show Me: **All Cases**

- Opened Date: **All Time**

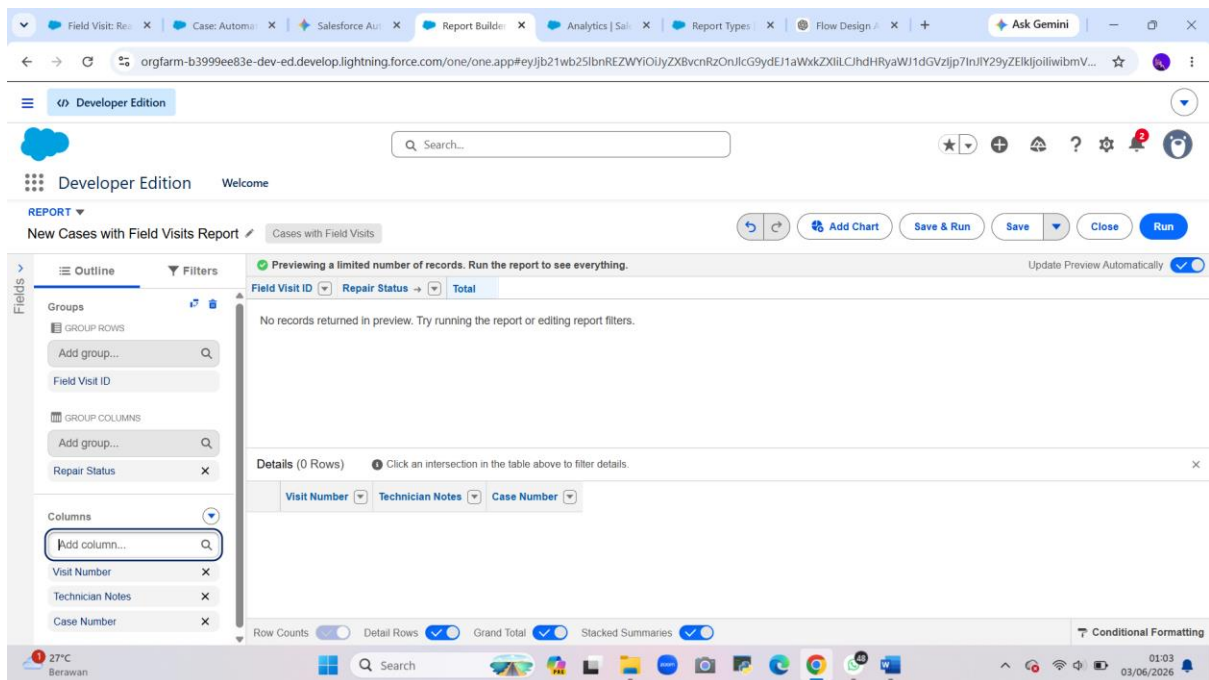


5. Under **Outline:**

- Group Rows by **Field Visit Owner**
- Group Columns by **Repair Status**

6. Add:

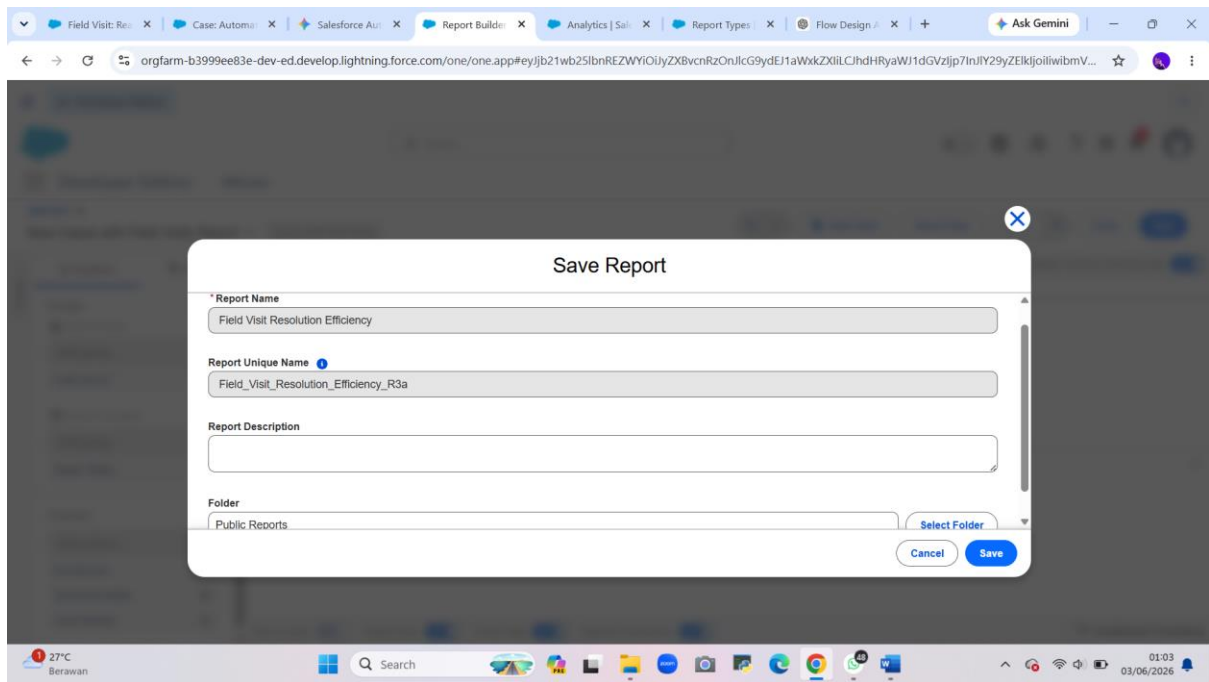
- Visit Number
- Technician Notes



7. Click **Save & Run**.

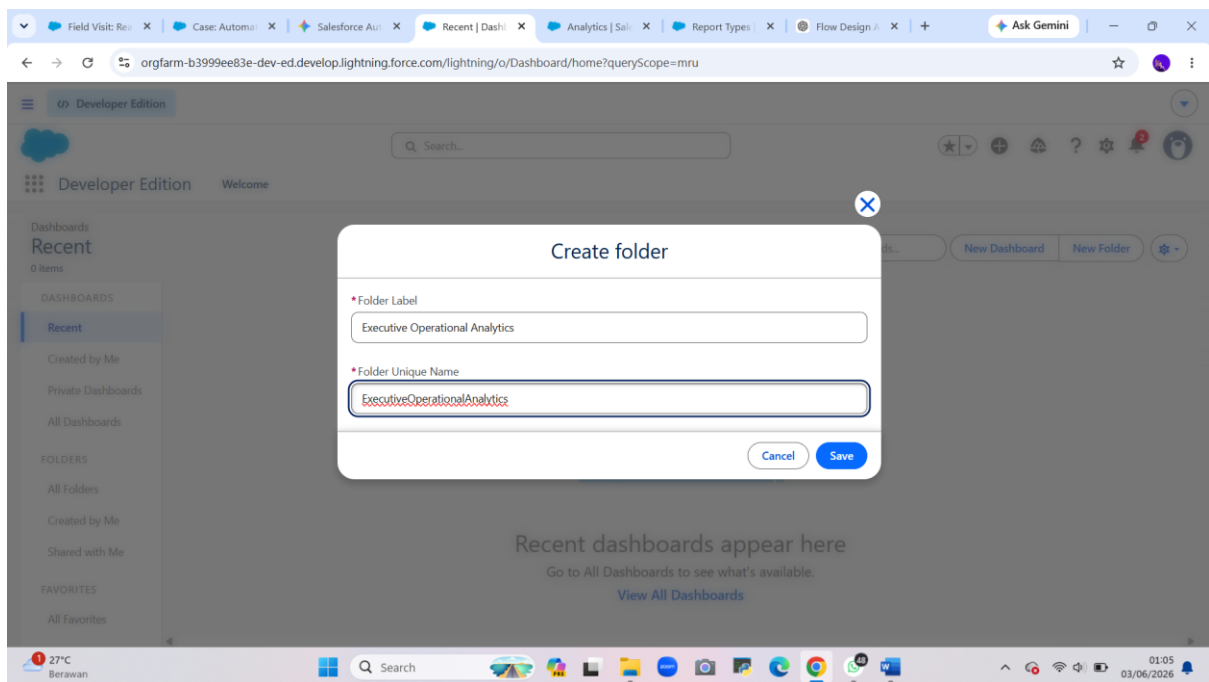
8. Save as:

- Report Name: Field Visit Resolution Efficiency
- Unique Name: Field_Visit_Resolution_Efficiency



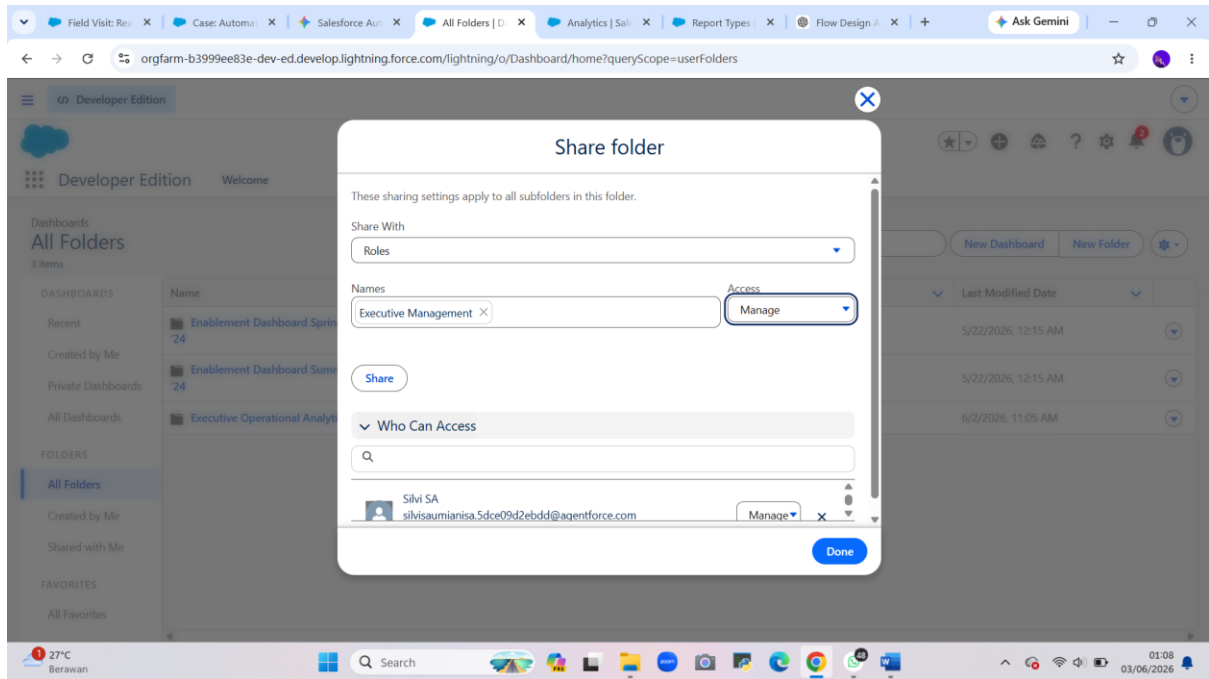
Step 4: Provision Secured Analytics Folders

1. Open the **Dashboards** tab.
2. Click **New Dashboard Folder**.
3. Name the folder:
 - Executive Operational Analytics

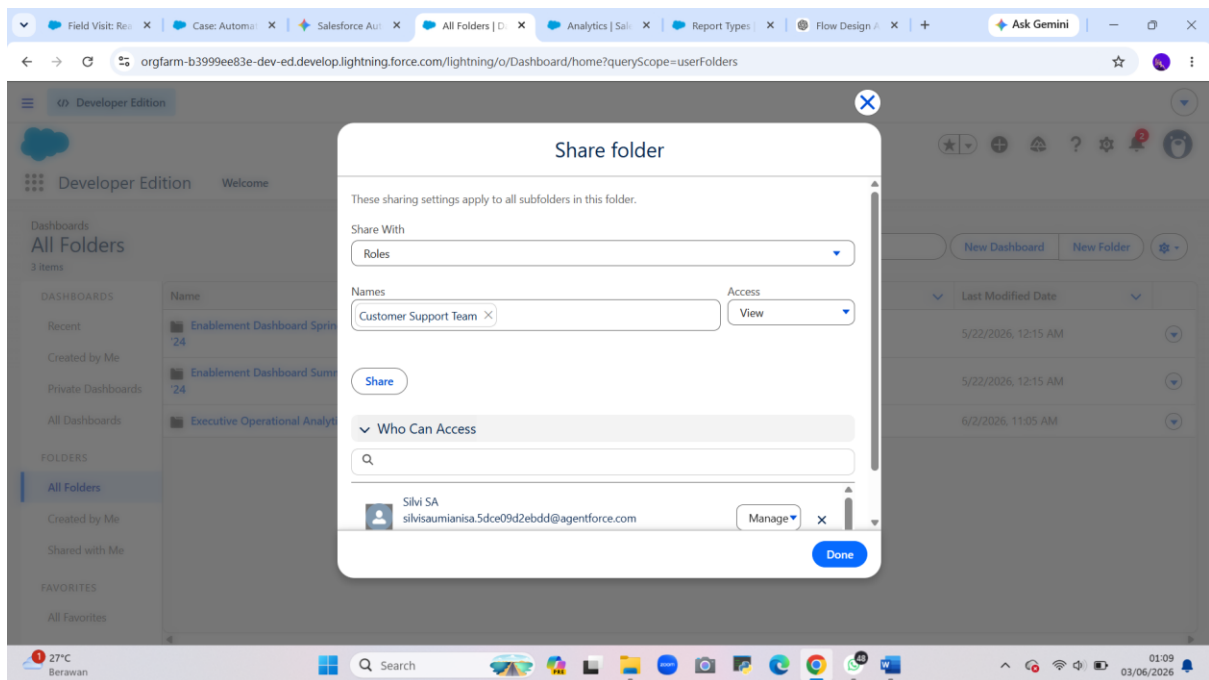


4. Click **Save**.

5. Open **Share** settings for the folder.
6. Configure:
 - Executive Management → Manage



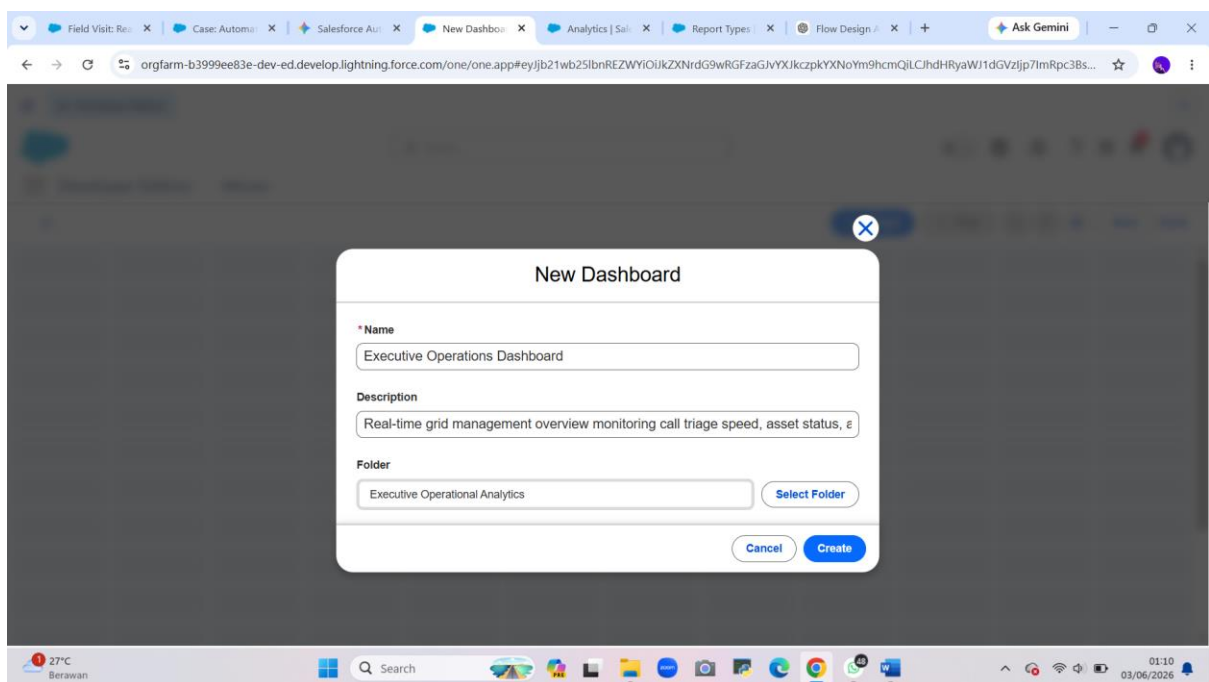
- Customer Support Team → Viewer



7. Click **Done**.

Step 5: Assemble the Executive Operations Dashboard

1. Open **Dashboards**.
2. Click **New Dashboard**.
3. Configure:
 - Name: Executive Operations Dashboard
 - Description: Real-time grid management overview monitoring call triage speed, asset status, and mobile technician field repair rates.
 - Folder: Executive Operational Analytics



4. Click **Create**.

Step 6: Add Component 1 — Live Open Crisis Counter

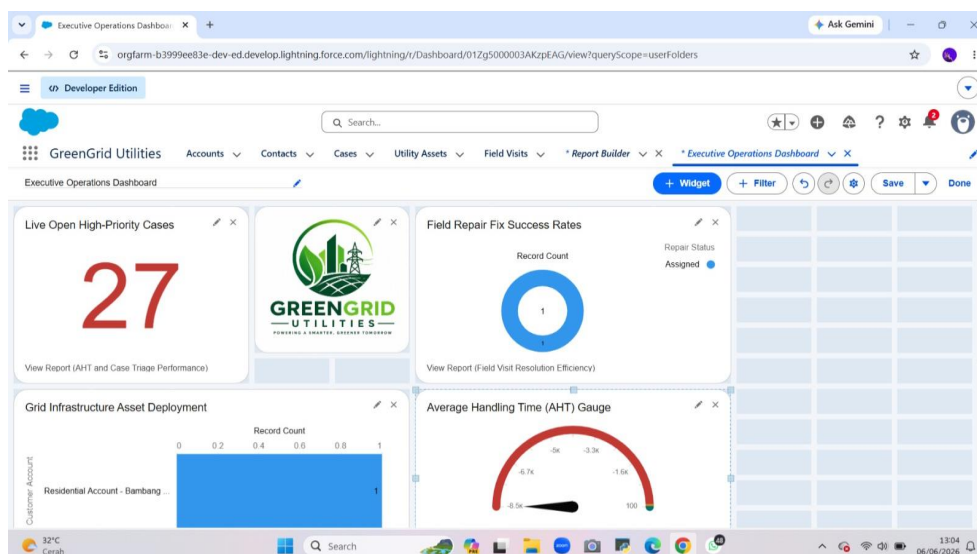
1. Click **+ Component**.
2. Select report:
 - AHT and Case Triage Performance
3. Configure:
 - Component Type: **Metric**
 - Measure: **Record Count**
 - Title: Live Open High-Priority Cases
4. Click **Add** and place it in the upper-left section.

Step 7: Add Component 2 — Service Console Speed Gauge

1. Click **+ Component**.
2. Select report:
 - AHT and Case Triage Performance
3. Configure:
 - Component Type: **Gauge**
 - Measure: **Average Age**
 - Breakpoint 1: 2
 - Breakpoint 2: 5
 - Title: Average Handling Time (AHT) Gauge
4. Click **Add** and position it in the top center.

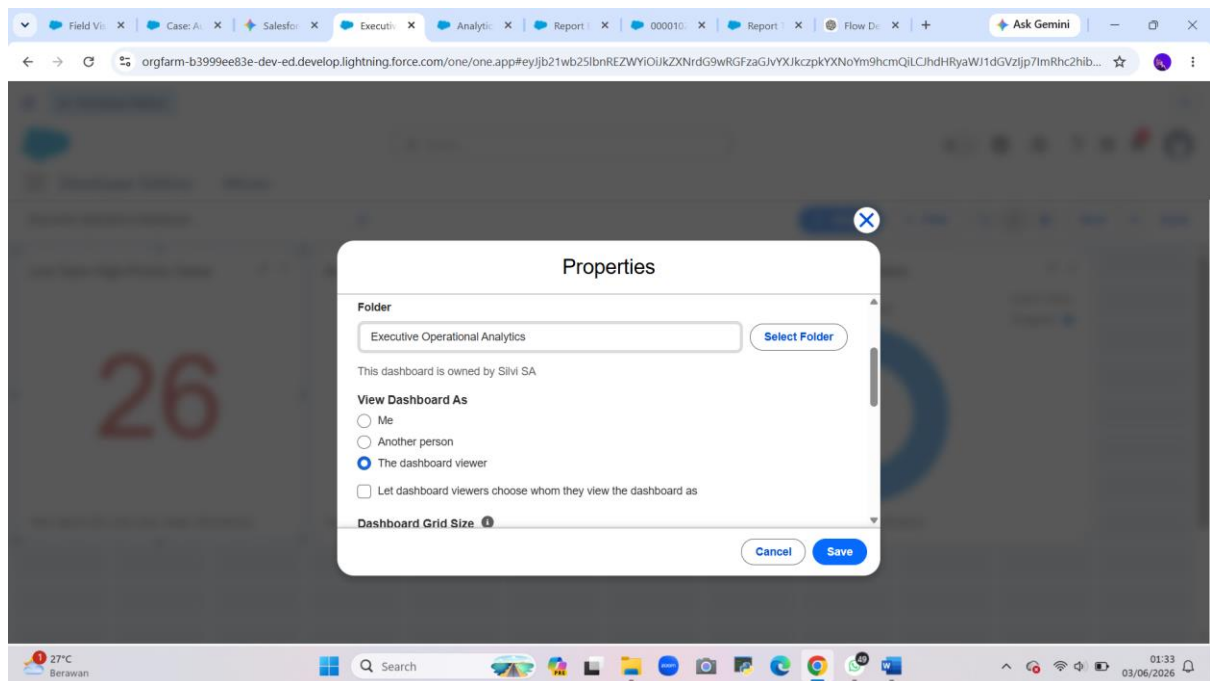
Step 8: Add Component 3 — First-Visit Fix Ratio Donut

1. Click **+ Component**.
2. Select:
 - Field Visit Resolution Efficiency
3. Configure:
 - Component Type: **Donut Chart**
 - Metric: **Record Count**
 - Slice By: **Repair Status**
 - Title: Field Repair Fix Success Rates
4. Click **Add** and position it in the upper-right section.



Step 9: Configure Dashboard Run Context and Finalize Activation

1. Open **Dashboard Settings**.
2. Under **View Dashboard As**:
 - Choose **The Dashboard Viewer** or
 - Select an Executive Management user.



3. Click **Save**.
4. Save the dashboard.
5. Click **Done** to launch the active dashboard.

Conclusion

With the completion of **Milestone 7: Reporting and Analytics Design**, GreenGrid Utilities has established a data-driven layer over its CRM architecture.

By mapping custom object relationships (Utility_Asset__c and Field_Visit__c) into structured reports and real-time dashboard metrics, the organization can track service speed gains, verify the reduction in handling times, and monitor field technician efficiency. Secure analytics folders ensure that sensitive operational performance metrics are delivered safely and efficiently to executive leadership.